

Lab 6 - Power Transformation

June 7, 2025

1 Theory

Power law (gamma) transformations are widely used in image processing for contrast enhancement. The general form of the power law transformation is:

$$S = C r^{\gamma}$$

where:

- S is the output pixel value,
- r is the input pixel value (normalized between 0 and 1),
- C is a constant (often set to 1),
- γ (gamma) is the exponent that controls the transformation.

By varying the value of γ , we can control how the input intensities are mapped to output intensities:

- If $\gamma < 1$, the transformation maps a narrow range of dark input values into a wider range of output values, making the image appear lighter.
- If $\gamma > 1$, the transformation maps a wide range of dark input values into a narrower range, making the image appear darker.
- When $\gamma = 1$, the transformation is linear and the output is identical to the input.

This family of transformations is useful for correcting the brightness and contrast of images, especially when dealing with images that are either too dark or too bright.

```
[4]: import numpy as np
      from PIL import Image
      import matplotlib.pyplot as plt
```

```
[5]: def power_function(input_image):
      img = input_image.resize((400,400), Image.Resampling.LANCZOS)

      # convert to numpy array
      numpy_image = np.array(img)
      numpy_image = numpy_image/255

      #transforming for different values of gamma
```

```

#plotting input and output images
# set up side-by-side image display
fig = plt.figure()
fig.set_figheight(150)
fig.set_figwidth(150)

fig.add_subplot(16,1,1)
plt.imshow(img, cmap='gray')
plt.title('original image')

gamma = 0.1
for i in range(15):
    #print(i*gamma)
    numpy_image_a = np.power(numpy_image,gamma*(i+1))
    numpy_image_a = numpy_image_a * 255
    numpy_image_a = np.around(numpy_image_a,decimals=0)
    power_image = Image.fromarray(numpy_image_a)
    power_image = power_image.convert("L")
    fig.add_subplot(16,1,i+2)
    plt.imshow(power_image, cmap='gray')
    plt.title('gamma = '+ str(round(i*gamma,1)))

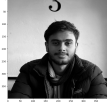
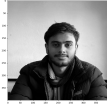
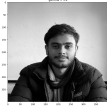
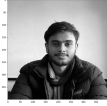
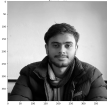
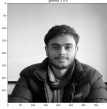
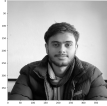
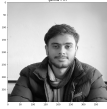
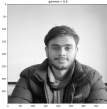
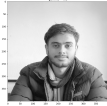
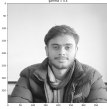
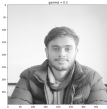
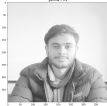
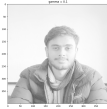
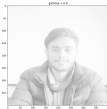
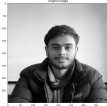
return power_image

```

```

[6]: # reading image and converting to gray scale
img = Image.open('img/samir.jpg').convert('L')
# Calling the power function
a = power_function(img)

```



2 Conclusion

From the lab work, we observe the effect of varying γ on a grayscale image. As γ increases from values less than 1 to values greater than 1, the output images transition from being lighter to darker. This demonstrates the flexibility of power law transformations in enhancing image details in different intensity ranges. By selecting an appropriate γ , we can improve the visual quality of images for better analysis and interpretation. This technique is fundamental in applications such as medical imaging, photography, and computer vision, where proper contrast adjustment is crucial.